

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off - Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2024

CC2103-1 – FUNDAMENTALS OF COMPUTER ARCHITECTURE AND ORGANIZATION
(For CCE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The partial remainder is restored by adding the divisor to the negative difference.
A) In non-restoration Division
B) In restoration method
C) In comparison method
D) In decimal arithmetic method
2. 2 x 2 bit array multiplier uses _____ AND gates and _____ half adders
A) 3 and 3
B) 4 and 2
C) 2 and 4
D) 2 and 2
3. _____ gives a procedure for multiplying binary integers in signed-2's complement representation
A) Booth algorithm
B) Multiplication algorithm
C) Comparison algorithm
D) Binary Search algorithm
4. _____ method uses the same address space for both memory and I/O.
A) isolated I/O
B) Memory-Mapped I/O
C) Interrupt-Driven I/O
D) Direct Memory Access (DMA)
5. The product of +7 and -11 is
A) 110011001
B) 110111000
C) 10110011
D) 111111100
6. _____ polynomial code obtained from the message bits by passing them through a feedback shift register containing a number of exclusive-OR gates, used for error-free messages.
A) Parity Check
B) Echo Check
C) longitudinal redundancy check (LRC)
D) Cyclic Redundancy Check (CRC)
7. A memory unit accessed by content is _____
A) Associative Memory
B) Virtual Memory
C) Primary Memory
D) Cache Memory
8. The _____ organization consists of a number of cross points that are placed at intersections between processor buses and memory module paths.
A) Time shared
B) Crossbar switch
C) DRAM
D) Multiport memory
9. A _____ is a storage device that stores information in such a manner that the item stored last is the first item retrieved
A) stack
B) Queue
C) Array
D) Tree

10. In CISC architecture, instructions are typically:
 A) Fixed-size B) Variable size
 C) Executed in one clock cycle D) Simple and require multiple clock cycles
11. The _____, referred to as reverse Polish notation (RPN)
 A) Infix notation B) Prefix notation
 C) Prepone notation D) postfix notation
12. _____ holds the address of the instruction to be executed next and is incremented each time an instruction is fetched from memory
 A) MAR B) PC
 C) MDR D) IR
13. A page fault can occur when:
 A) The page table is full B) The required page is not in main memory
 C) The cache is full D) The disk drive fails
14. The result of $+42+(-13)$ in binary is
 A) 00011101 11110011
 C) 00001101 11100011
15. Convert binary number 10010110 into octal number.
 A) 226 B) 224
 C) 221 D) 222
16. The primary purpose of an Operating system is
 A) To keep system programs employed B) To allow people to use the computer
 C) To make computers easier to use D) To make the most efficient use of computer hardware
17. 5BC Hexadecimal to the decimal system
 A) 1468 B) 1648
 C) 468 D) 1846
18. A _____ is a combinational circuit that receives binary information from one of 2^n input data lines and directs it to a single output line.
 A) Full – Adder B) Half- Adder
 C) NAND gate D) Multiplexer
19. A 4-bit register has a group of _____ flip-flops and is capable of storing any binary information of _____
 A) 3,2 B) 2, 3
 C) 1, 4 D) 4, 4
20. Perform the subtraction with the unsigned decimal number by taking the 2's complement of the subtrahend: $11010 - 10000$
 A) 10110 B) 11101
 C) 01010 D) 01101

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Explain with the neat functional block diagram the working of the computer.	10	L*2	1	2
	b) Perform the subtraction on the following pairs of numbers using 5-bit signed 2's complement format. Determine about overflow in each case i) +27 and +9 ii) -11 and -5 iii) +22 and -8	6	L3	2	1
2.	a) Compare and contrast single-bus and multiple-bus architectures. Discuss the advantages and disadvantages of each	10	L2	1	2
	b) Given the multiplicand A = +42 and multiplier B = -16. Solve the multiplication of A and B using Booth's algorithm.	6	L3	2	1,2,3

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|-------|---|----|----|---|---|
| 3. a) | Explain with the neat diagram the Multiplication of floating point numbers. | 10 | L2 | 1 | 1 |
| b) | Explain the Integer representation of signed numbers. Convert the following decimal numbers to the bases indicated. | | | | |
| a. | 7562 to octal | 6 | L3 | 1 | 2 |
| b. | 1938 to hexadecimal | | | | |
| c. | 175 to binary. | | | | |

Unit – II

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|-------|--|----|----|---|-----|
| 4. a) | Given that registers R3 and R4 contain the decimal values 2400 and 5800 respectively, what is the effective address of the source or destination operand in each of the following instructions? | | | | |
| i. | Load 50(R3), R6 | | | | |
| ii. | Store R6, 40(R3,R4) | | | | |
| iii. | Add (R4)+, R6 | 6 | L2 | 2 | 1,2 |
| b) | Explain the phases of the instruction cycle, including fetching, decoding, executing, and storing. How do these phases vary depending on the type of instruction being processed? | 10 | L2 | 2 | 1,2 |
| 5. a) | Describe the interrupt handling process. A print routine is stored in memory starting at address 3000, and it is executed when the computer identifies an interrupt triggered by INTR = 1 (with EN = 1). | | | | |
| i. | What instruction should be placed at address 0? | 6 | L2 | 3 | 1,8 |
| ii. | What should the last two instructions of the print routine be? | 10 | L3 | 2 | 3 |
| b) | Explain 4-bit arithmetic circuit with a neat diagram and Example | | | | |
| 6. a) | Compare and contrast the following | | | | |
| i. | Hard-wired control | | | | |
| ii. | Micro programmed control | 6 | L2 | 3 | 3,1 |
| b) | Demonstrate DMA and its implementation and show how the data is transferred between memory and I/O devices using the DMA controller. | 5 | L2 | 3 | 1,2 |
| c) | Describe how function calls and returns are managed in a stack-based organization. What role does the stack play in handling local variables and control information? | 5 | L2 | 3 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Explain the concept of cache coherence in multiprocessor systems. What are the challenges associated with ensuring cache coherence? | 6 | L2 | 5 | 3,1 |
| b) | Define shift microoperations and explain the different types of shifts (logical, arithmetic, and circular) used in digital systems. | 5 | L2 | 4 | 3,2 |
| c) | Explain with a Flowchart of interrupt cycle. | 5 | L2 | 5 | 3,2 |
| 8. a) | Define direct mapping in the context of cache memory. How does it differ from associative and set-associative mapping? | 6 | L2 | 4 | 3,2 |
| b) | Explain the concept of RAID (Redundant Array of Independent Disks). Discuss any 2 RAID levels and their trade-offs in terms of performance, data redundancy, and cost. | 5 | L2 | 5 | 3,2 |
| c) | Define pipelining in the context of computer architecture. How does pipelining improve the performance of a CPU? | 5 | L2 | 5 | 3,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
